Three MIMIC-III Predictions, One Design Discipline

Drug recommendation, next-visit mortality and readmission
built by letting the data choose the architecture.

Janak Sharma · **Alex Morman** · **Andy Trinh**

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Why Does it Matter?

Drug Recommendation

Intelligent recommendations can support clinicians in making safer, personalized treatment decisions.

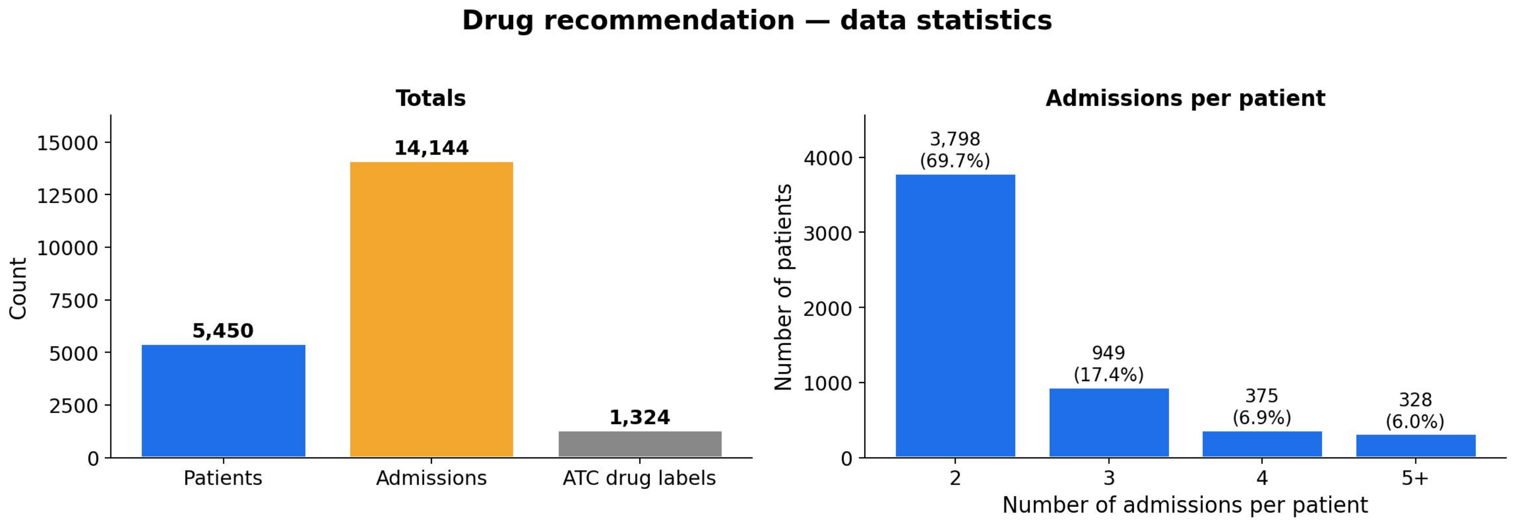
Mortality Prediction

Identifying high-risk patients can improve health outcomes

Readmission Prediction

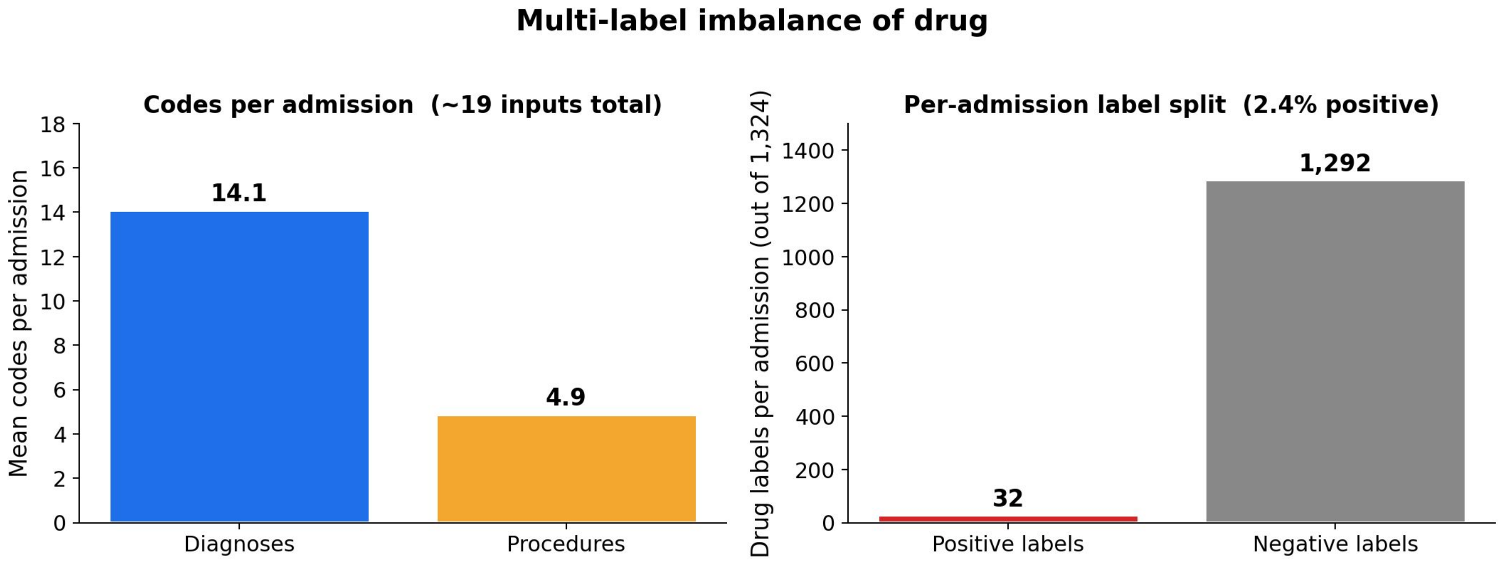
Helps administrators manage their limited resources by predicting readmission of discharged patients

What Each Task Looks Like in the Data



Patients have fewer history

- around 70% of patients got only 2 visits



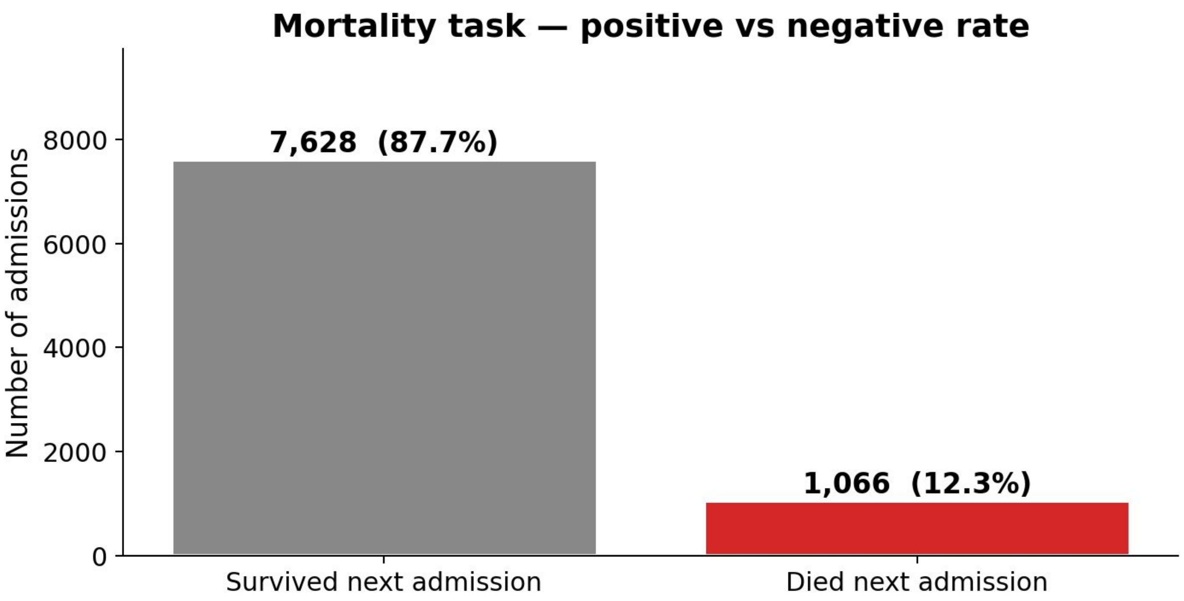
Class Imbalance

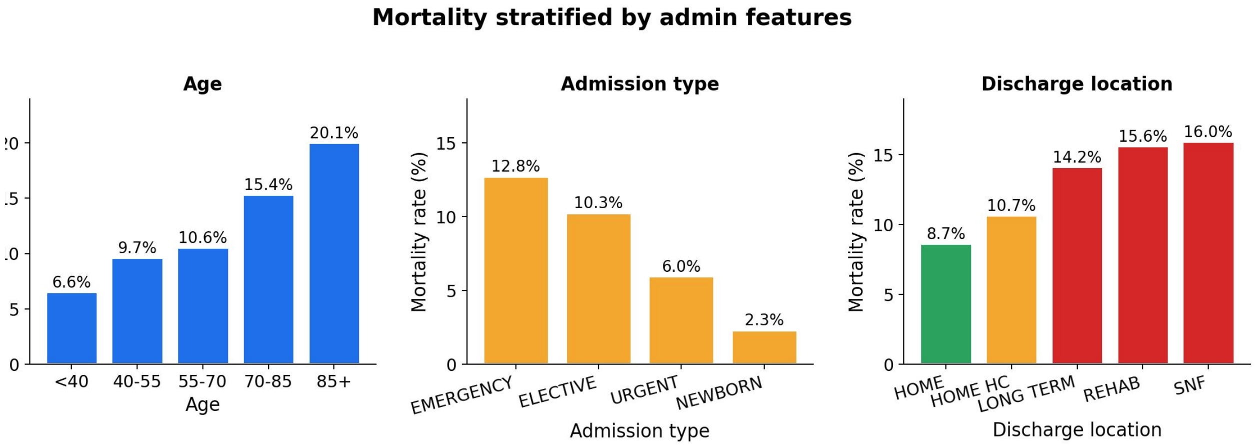
- out of 1324 total drugs only 32 are prescribed on an average

Visit has information

- single visit contains diagnoses, procedures of around 19 on an average

What Each Task Looks Like in the Data



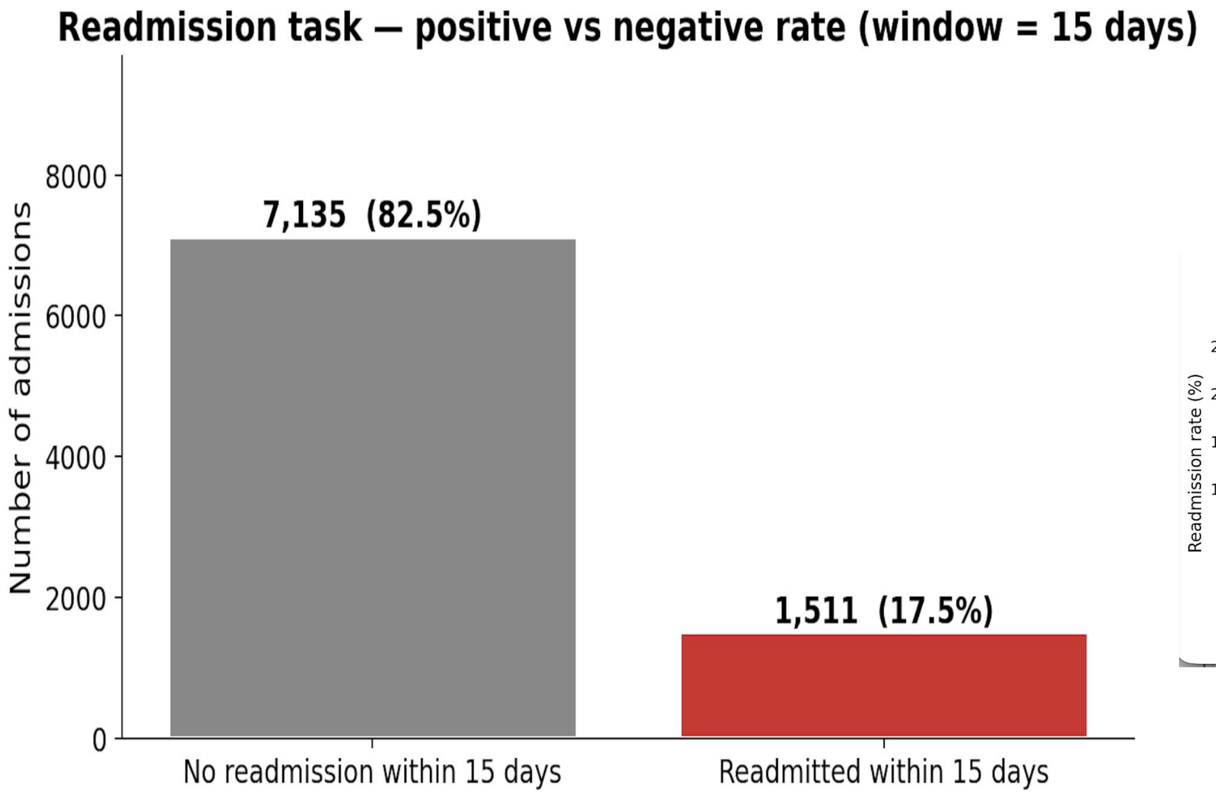


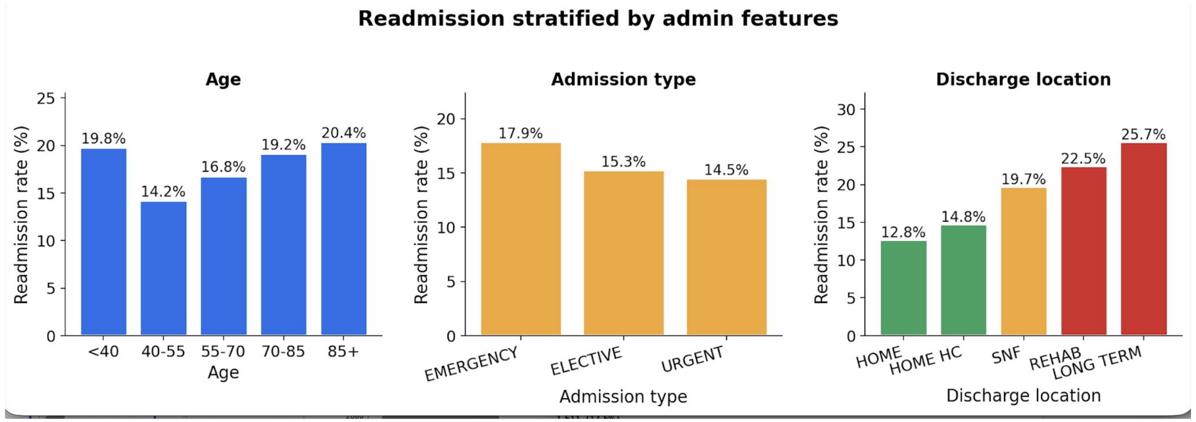
Mortality — age, admission type, discharge location

Mortality rises 6.6% (<40) ⟶ 20.1% (85+) by age, and 8.7% (HOME) ⟶ 16.0% (SNF) by discharge location.

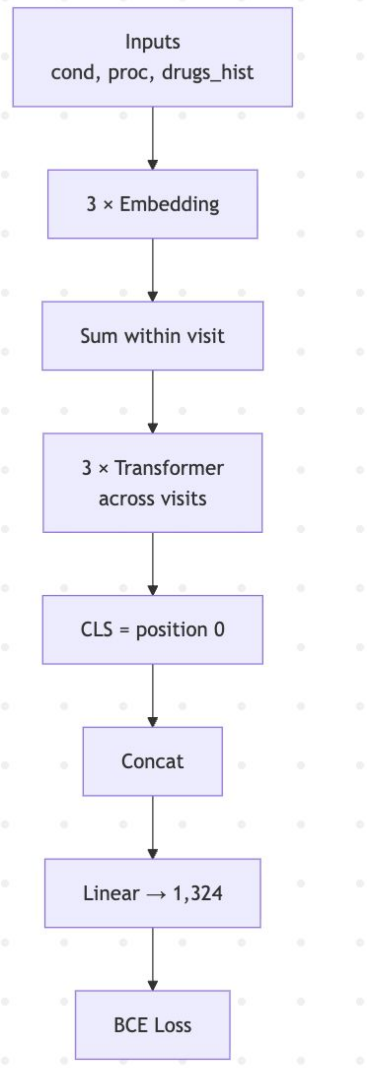
Three admin signals — age, admission type, discharge location are correlated to death

What Each Task Looks Like in the Data

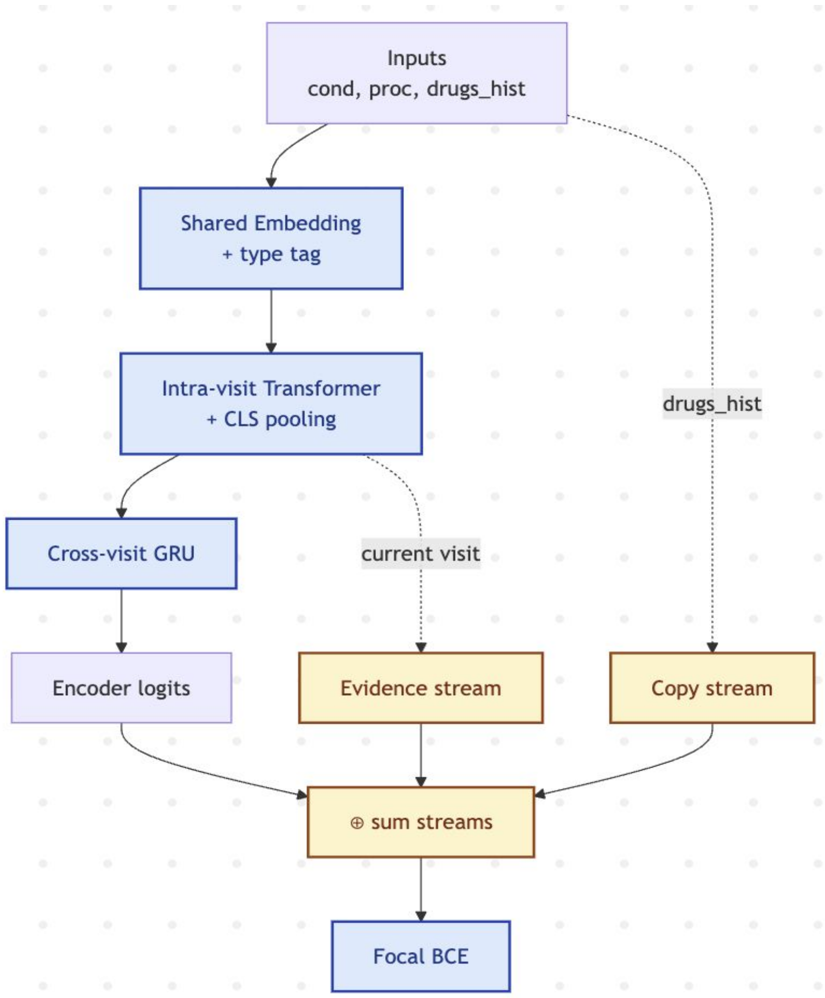




HCAT: Drug Recommendation

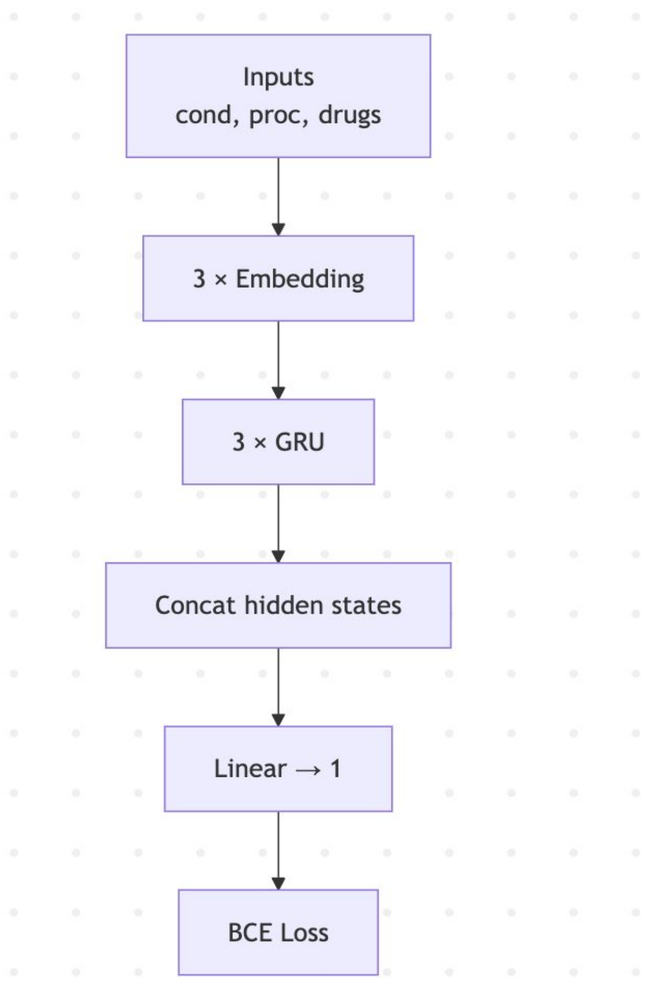


Baseline Architecture

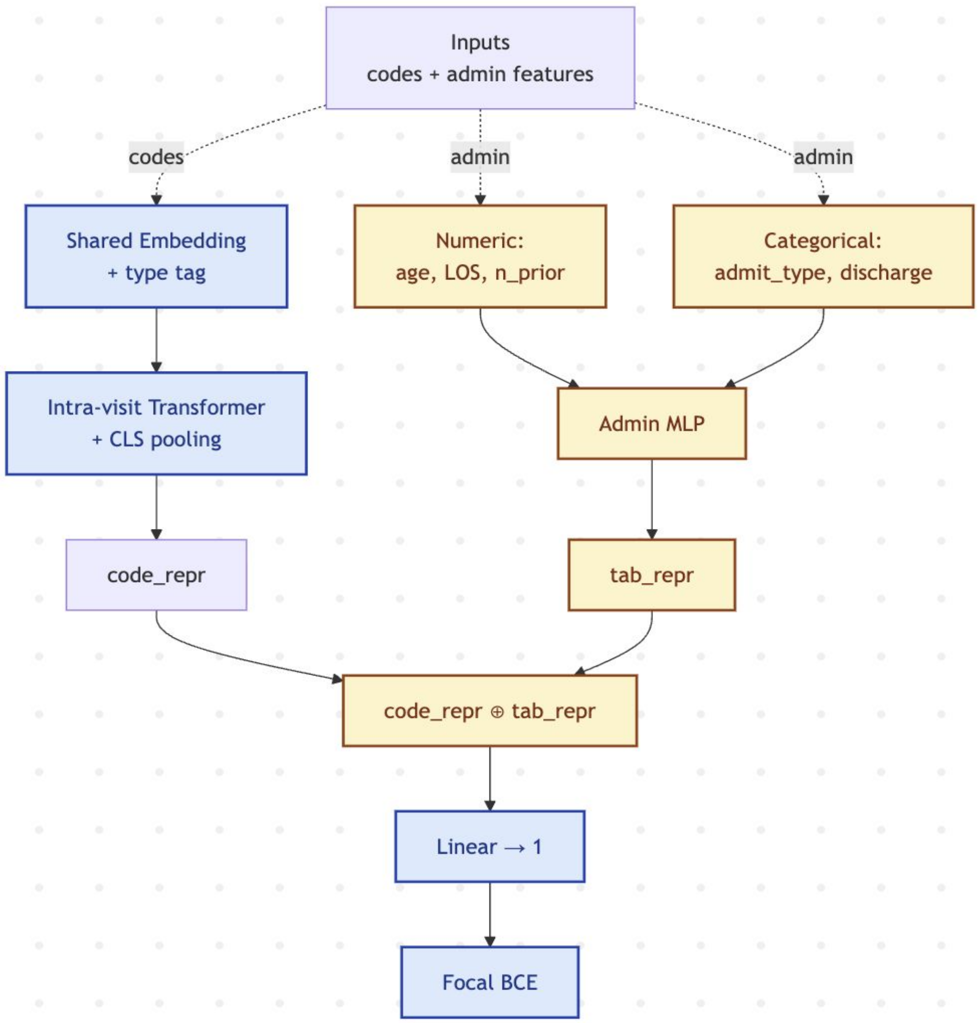


Proposed Architecture

HCAT: Mortality Prediction & Readmission Prediction



Baseline Architecture



Proposed Architecture

Training Hyperparameters

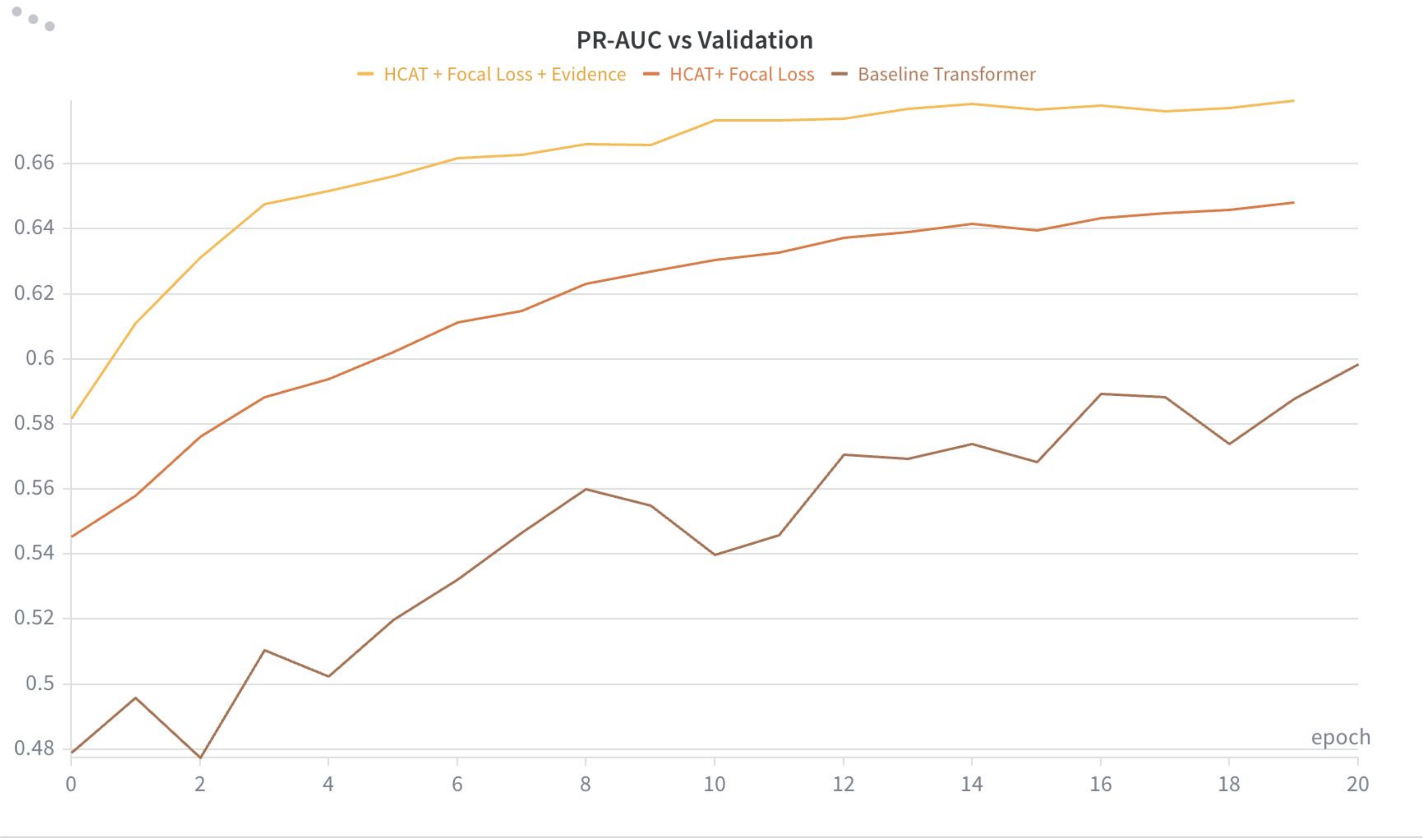
Hyperparameters	Drug Rec	Mortality/Readmission
Embedding dim	128	128
Intra-visit Transformer	1 layer, 2 heads	1 layer, 2 heads
Cross-visit module	1-layer GRU	— (single-visit task)
Admin / structured path	—	MLP (Linear → GELU → LN → Linear), 3 numeric + 2 categorical features
Logit fusion	encoder + copy + evidence	code_repr + tab_repr
Loss	Focal BCE (γ=2)	Focal BCE (γ=2)
Optimizer	Adam, lr=1e-3, wd=1e-4	Adam, lr=1e-3, wd=1e-4
Batch / Epochs	32 / 20	32 / 20
Total params	~1.71 M	~1.25 M
Output	1,324 drug logits	1 logit

Drug Recommendation: Improvements over Baseline

Δ (green) = method − baseline.   Highlighted row = primary metric.

Metric	Baseline (PyHealth Transformer)	Variant 1 (Hierarchical Transformer + Focal BCE)	Δ vs Baseline	Variant 2 (Hierarchical + Focal + Copy/Evidence)	Δ vs Baseline
AUPRC	**0.6062 ± 0.004**	**0.6424 ± 0.003**	**+0.0362 (+5.97%)**	**0.6734 ± 0.005**	**+0.0672 (+11.08%)**
Precision@10	0.7817 ± 0.006	0.8114 ± 0.004	**+0.0297 (+3.80%)**	0.8266 ± 0.007	**+0.0449 (+5.74%)**
Recall@10	0.2472 ± 0.002	0.2576 ± 0.002	**+0.0104 (+4.21%)**	0.2646 ± 0.002	**+0.0174 (+7.04%)**
Precision@20	0.6770 ± 0.004	0.7087 ± 0.003	**+0.0317 (+4.68%)**	0.7307 ± 0.005	**+0.0537 (+7.93%)**
Recall@20	0.4163 ± 0.002	0.4366 ± 0.003	**+0.0203 (+4.87%)**	0.4544 ± 0.003	**+0.0381 (+9.15%)**

Drug Recommendation: Improvements over Baseline

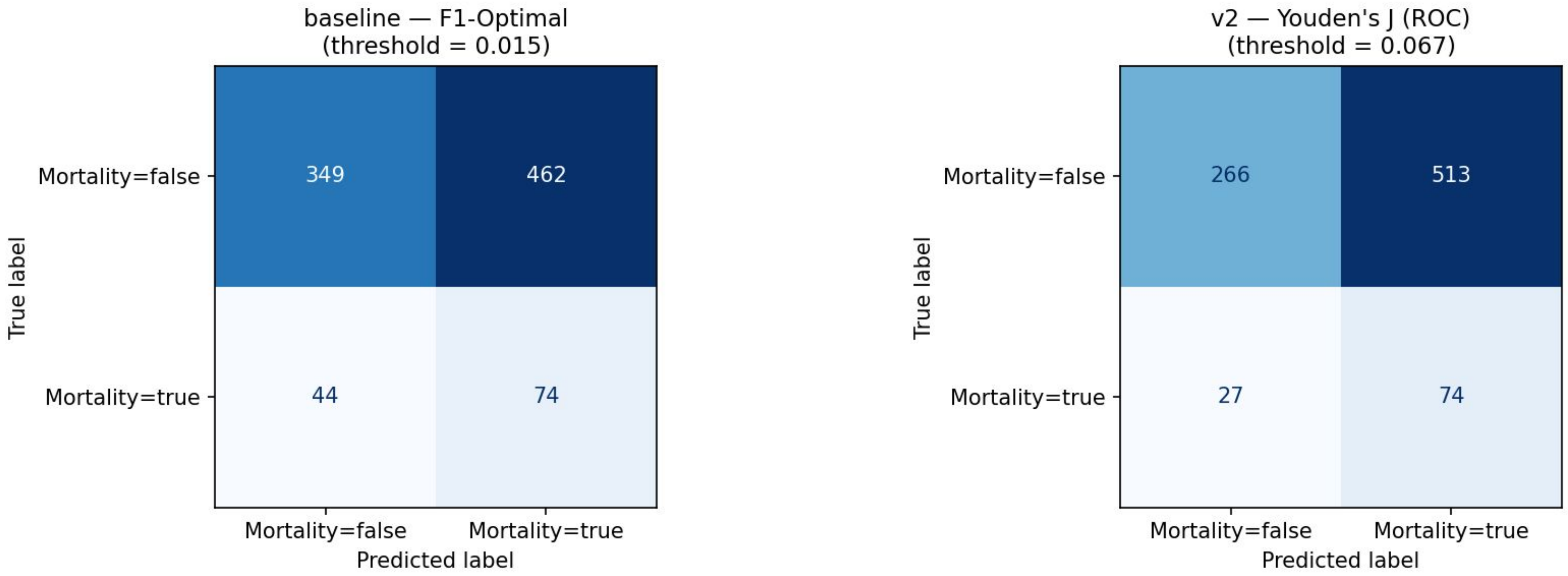


Mortality Prediction: Improvements over Baseline

Δ (green) = method − baseline.   Highlighted row = primary metric.

Metric	Baseline (PyHealth RNN)	HCAT	Δ vs Baseline	HCAT (Codes Ablated)	Δ vs Baseline	HCAT (Admissions Ablated)	Δ vs Baseline
AUROC	**0.6013 ± 0.0233**	**0.6239 ± 0.0224**	**+0.0226 (+3.76%)**	**0.6329 ± 0.0337**	**+0.0316 (+5.25%)**	**0.5676 ± 0.0383**	**-0.0337 (-5.61%)**
AUPRC	0.1828 ± 0.0228	0.2253 ± 0.0200	**+0.0424 (+23.22%)**	0.2143 ± 0.0329	**+0.0314 (+17.19%)**	**0.1516 ± 0.0206**	**-0.0312 (-17.07%)**

Mortality Prediction: Improvements over Baseline



Recall increases greatly

Slight precision penalty

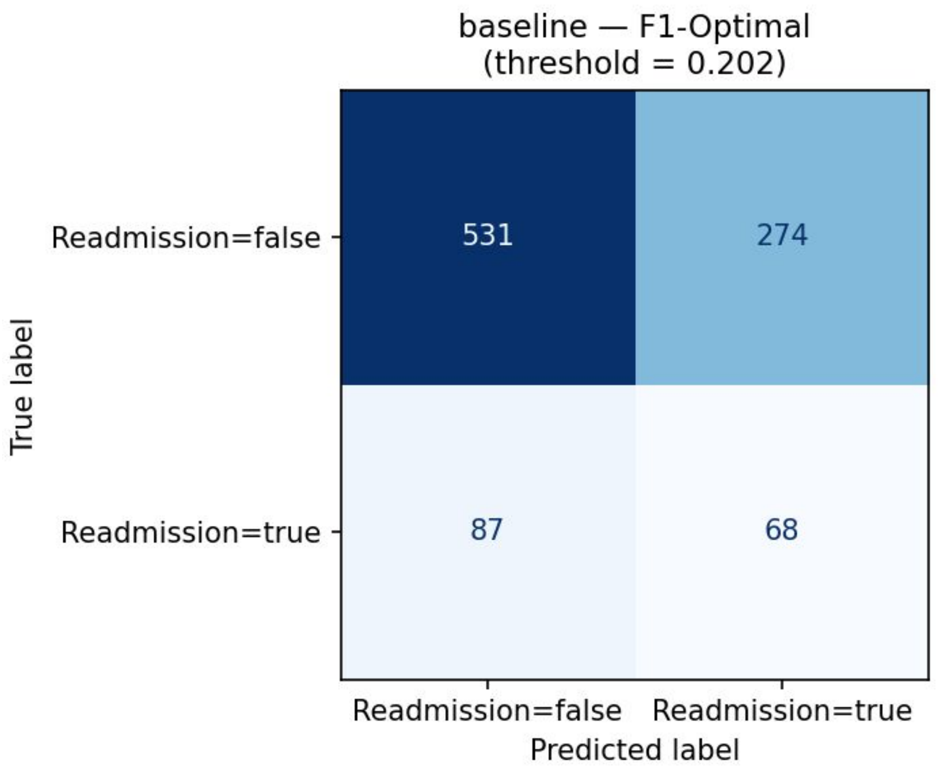
We used Youden's J [3] to determine an optimal threshold

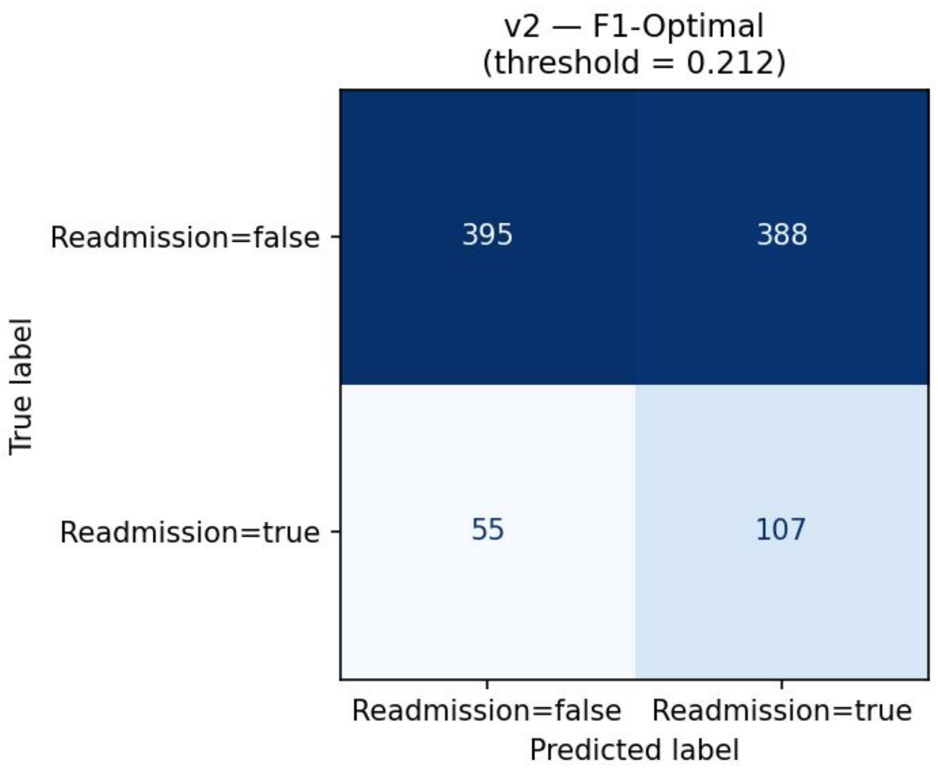
Readmission Prediction: Improvements over Baseline

Δ (green) = method − baseline.   Highlighted row = primary metric.

Metric	Baseline (PyHealth RNN)	HCAT	Δ vs Baseline	HCAT (Codes Ablated)	Δ vs Baseline	HCAT (Admissions Ablated)	Δ vs Baseline
AUROC	**0.5927 ± 0.0236**	**0.6151 ± 0.0234**	**+0.0224 (+3.78%)**	**0.6014 ± 0.0121**	**+0.0087 (+1.47%)**	**0.5730 ± 0.0390**	**-0.0196 (-3.31%)**
AUPRC	0.2426 ± 0.0246	0.2596 ± 0.0329	**+0.0170 (+7.02%)**	0.2579 ± 0.0217	**+0.0153 (+6.32%)**	**0.2273 ± 0.0346**	**-0.0153 (-6.31%)**

Readmission Prediction: Improvements over Baseline





Recall increases greatly

Precision improves slightly

Discussions

What worked?

Data-driven design

Every component traces back to a measurable property of the data.

Additive logit streams (copy path)

Cleanly handle cold-start vs with-history without branching code.

Change of Loss Function

handled class imbalance probem

Intra Visit Transformer and GRU

Gave some gain compared to mean pooling of baseline

What didn’t work?

Complex Transformer

Adding extra layers and heads to the transformer

Increasing embedding size may be due to datasize

Changing Embedding model to BioBERT

convergence was seen in few epochs but training was slow

No significant improvement in score

Admin signals

adding admin signals didn’t improved drug recommendation

Limitations

Codes were ineffective

Other admin signals were showing stronger correlation to death than the ICD Code signals

Data Distribution

Highly imabalanced dataset makes reliable and accurate inference difficult.

Hardware limitations

The LABEVENTS table was too heavy to process quickly on available consumer grade hardware. It contained potentially useful data, particularly the abnormal flag, but we did not have the training time to utilize it

Conclusion, Takeaways & Future Work

Real World EHR Data is complex

Real-world data is complex and noisy. It isn’t like Kaggle datasets where every relationship is nice, neat and easy

Model Interpretation Analysis

Our results are still fresh, and we haven’t had a great opportunity to analyze the models at the component level.

Consider Additional Data Tables

Tables like NOTE_EVENTS, LAB_EVENTS, and others contain information which might improve model performance.

Questions?

Janak Sharma · Alex Morman · Andy Trinh

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References

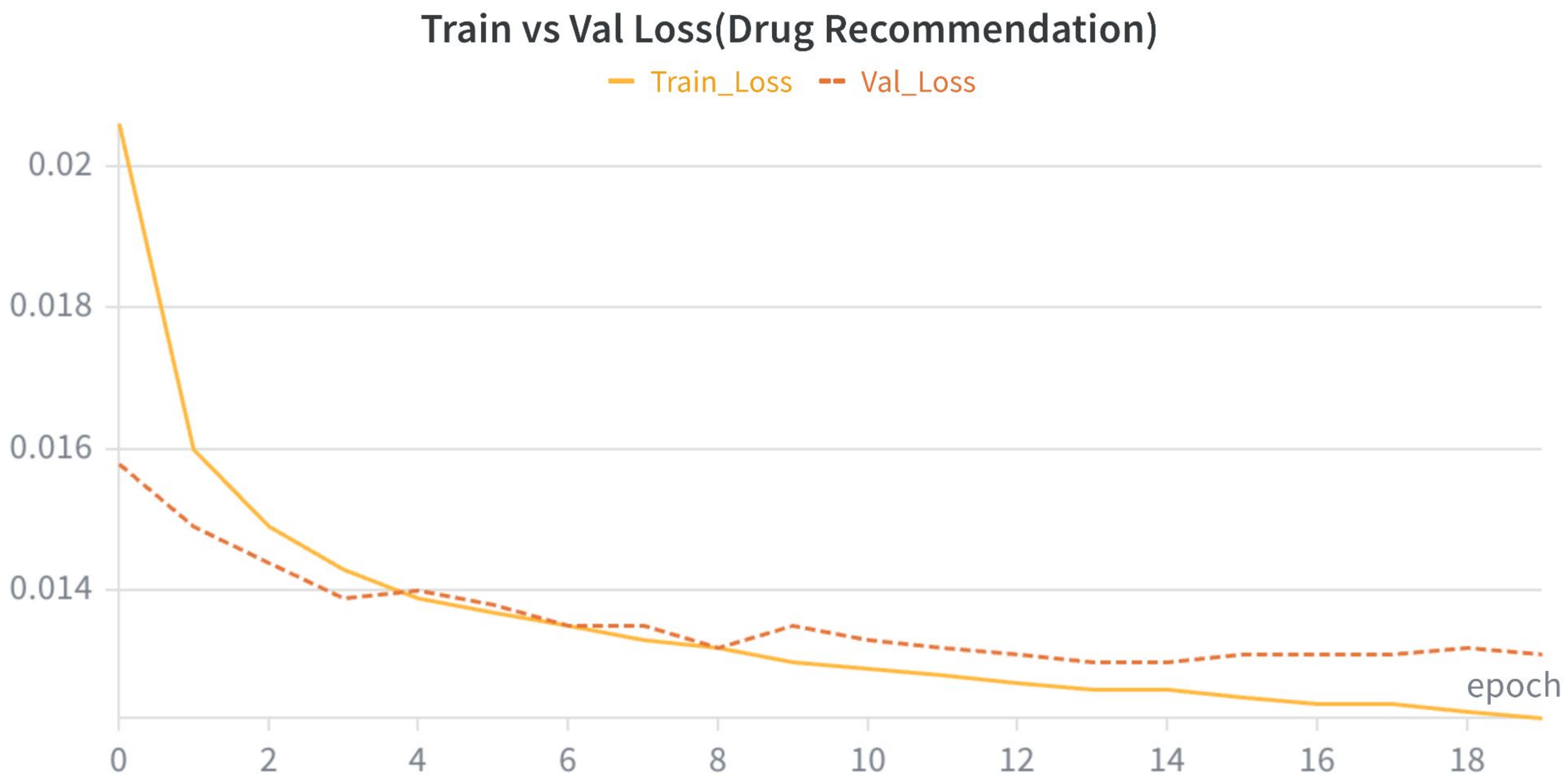
[1] A. E. W. Johnson, T. J. Pollard, L. Shen, L. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. A. Celi, and R. G. Mark, "MIMIC-III, a freely accessible critical care database," *Scientific Data*, vol. 3, p. 160035, 2016. doi: 10.1038/sdata.2016.35.

[2] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, Venice, Italy, Oct. 2017, pp. 2980–2988. doi: 10.1109/ICCV.2017.324.

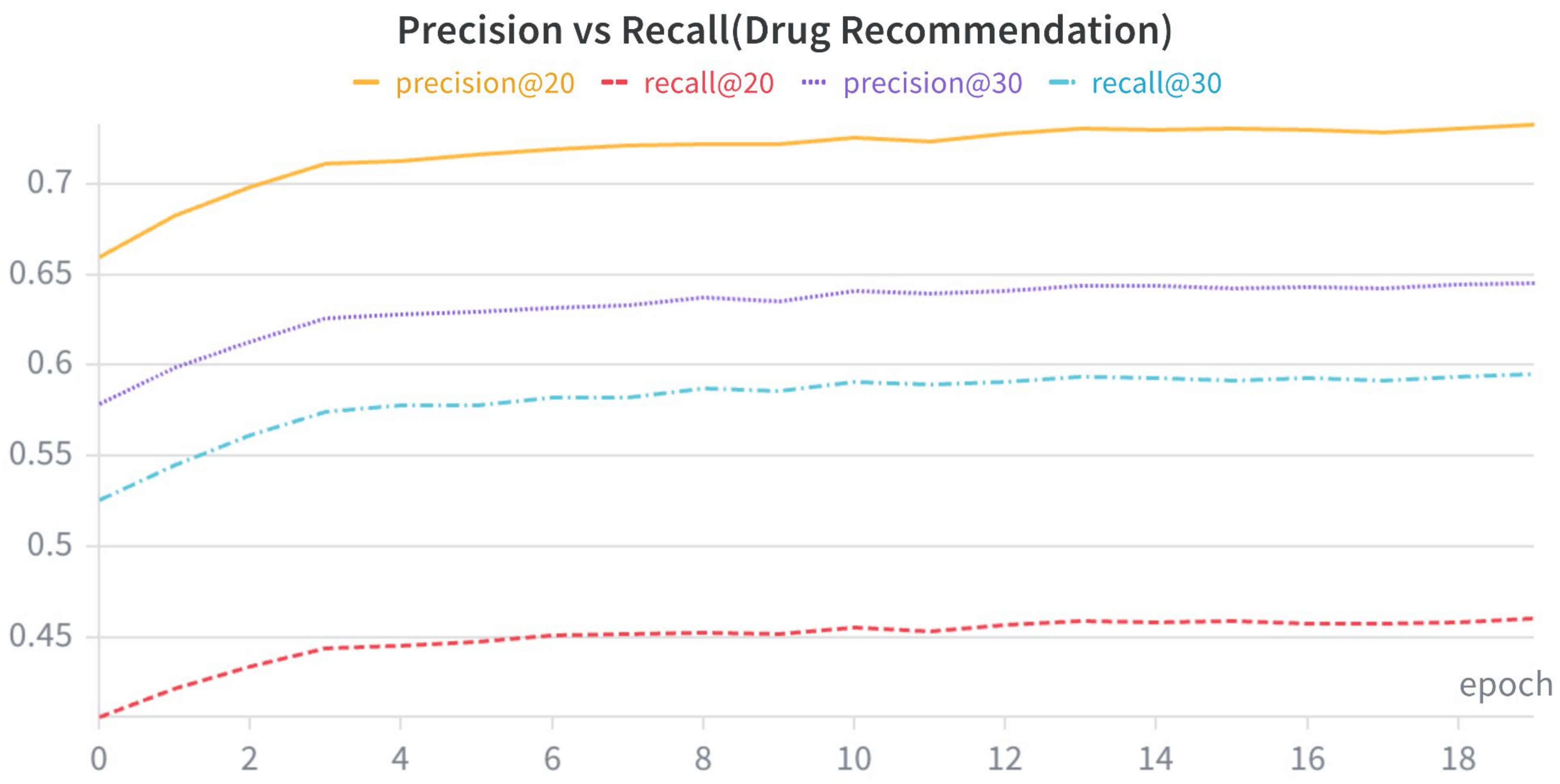
[3] W. J. Youden, "Index for rating diagnostic tests," *Cancer*, vol. 3, no. 1, pp. 32–35, 1950. doi: 10.1002/1097-0142(1950)3:1<32::AID-CNCR2820030106>3.0.CO;2-3.

Appendix: Bonus Visualizations Ahead

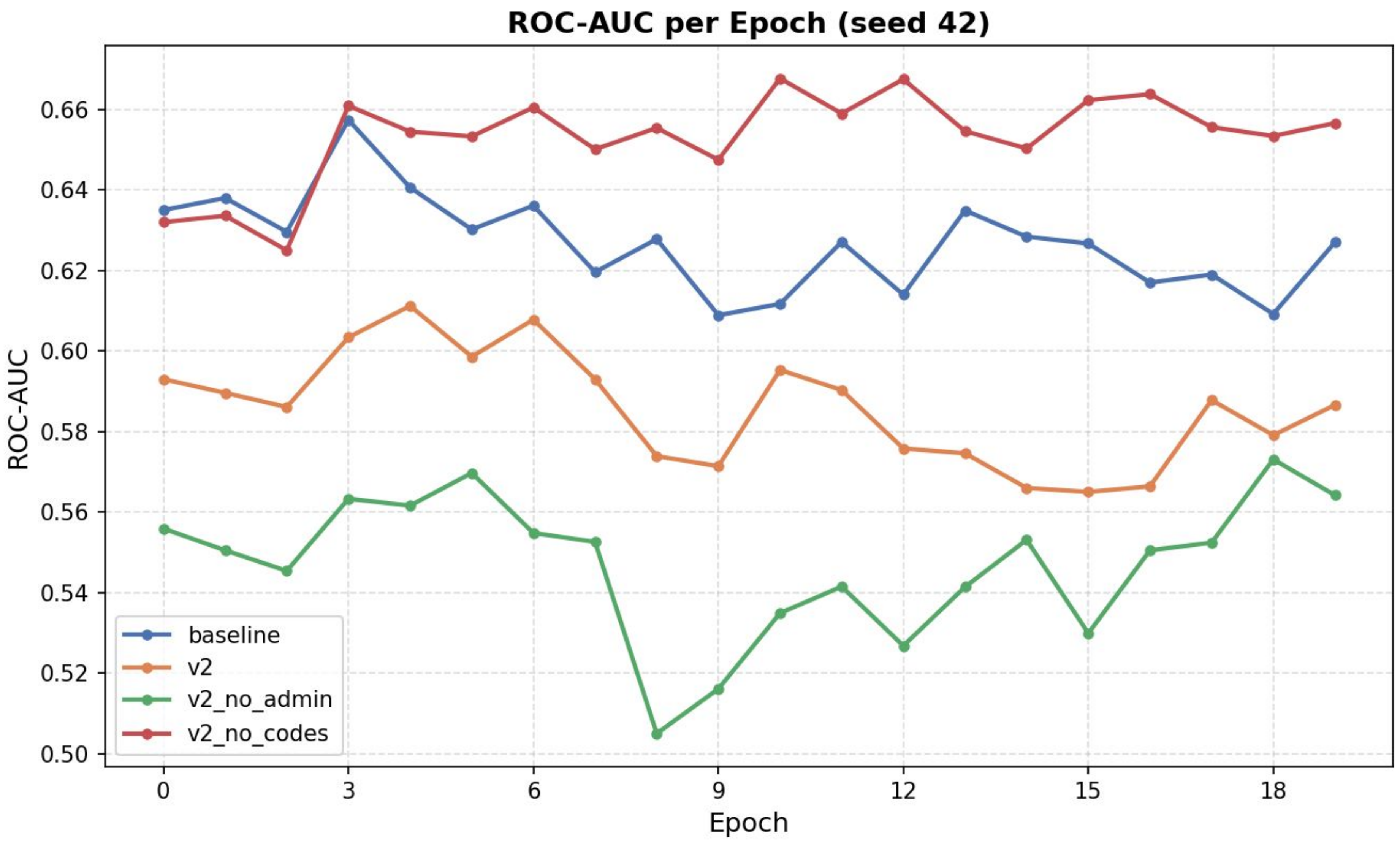
Drug Recommendations



Drug Recommendations



Mortality Prediction: Improvements over Baseline



Readmission Prediction — Improvements over Baseline